

AROCHA, VANESSA ANN A.

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Educational Background

Mindanao State University - Iligan Institute of Technolog
Bachelor of Science in Computer Application

Major in Embedded System
(UnderGrad)

Iligan City National Highschool
2019 - 2021
Mahayahay, Iligan city
With Honor

Hinaplanon National Highschool
2018 - 2019
Cabughason, Hinaplanon, I.C.

Villanueva National Highschool
2016 - 2018
Looc, Villanueva Mis.Or.

Looc Elementary School
2009 - 2015
Looc, Villanueva Mis.Or.

Language

- English
- Filipino
- Bisaya

Skills

- **Programming Languages:**
  C, C++, PHP, Python, Assembly Language, JavaScript, SQL
- **Web & Database:**
  HTML5, CSS3, MySQL, PHPMyAdmin
- **IoT & Hardware:**
  ESP32, ESP32S3, Arduino, RFID Technology, Circuit Design, Power Supply Systems, Raspberry Pi 4, Programmable Logic Controller
- **Development Tools:**
  VS Code, Arduino IDE, Qt Creator, Cisco Packet Tracer, Google Colab

PROFILE

A dedicated BS Computer Applications senior majoring in Embedded Systems, hoping to gain hands-on experience through an OJT position at ILPI. Familiar with low-level programming in C, C++, and Python. Eager to learn, open to guidance, and committed to supporting the team in managing information systems efficiently.

ACADEMIC PROJECTS

B.B.S : Baranggay Blotter System **SY 2022-2023 SEM 2**

Computer Programming 2

- Designed a system to digitize and streamline the barangay blotter/complaint process
- Enabled efficient storage and retrieval of complaint records
- Automated generation of complaint documents
- Implemented tracking to identify repeated complaint attempts by the same individuals

Power Supply **SY 2022-2023 SEM 2**

Fundamentals of Computer Applications 2

- Built a stable AC-to-DC power supply using transformers, capacitors, and voltage regulators
- Assembled and troubleshot circuits for consistent output

Amplitude-Modulated Radio Transmitter **SY 2023–2024 SEM 1**

Advanced Computer Circuits

- Built a functional AM radio transmitter for short-range audio transmission
- Applied analog circuit design, signal processing, and modulation techniques

Integrated Security and Monitoring System for Sikyop **SY 2024-2025 SEM 1**

Programming For Internet of Things

- Built an environmental monitoring system using water level, raindrop, and PIR sensors around Sikyop
- Provided real-time data to support tourism management decisions

Banana Quality Inspection and Lifespan Detection Using Raspberry Pi Pico-Based Embedded System **SY 2025-2026 SEM 1**

Embedded System Programming

- Built a low-cost sensor system to track temperature, humidity, and methane gas for forecasting banana shelf life
- Improved post-harvest storage to reduce food waste
- Delivered affordable tech for farmers to support sustainable farming and food security

An Exploratory Design and Evaluation of a XIAO ESP32S3-Based Edge Computing Prototype for Huanglongbing (Citrus Greening) Detection in Calamansi **SY 2025-2026 SEM 2**

Undergraduate Thesis

- Built an edge computing prototype with the XIAO ESP32S3 to detect Huanglongbing (citrus greening disease) in calamansi
- Combined embedded systems and on-device processing for real-time detection
- Evaluated performance to support early diagnosis and better crop management